ASSIGNMENT 4.1

Belief-Desire-Intention (BDI) Goal-Driven Modelling

Student	[Student Name]	Roll No	[Roll No]
Course	21CSE525T - Agent Technology	Project	Placement Preparation Agent

Unit 4 - Deliberative architecture with dynamic plan recalculation. File: assignments/a4_1/bdi_agent.py

1. Objective

Model the agent as Belief, Desire and Intention components, and show it abandoning a committed plan when a change in belief makes a different goal more urgent.

2. Components

- **BeliefBase:** facts about the student and the world (for example days_to_drive, weakest_topic). Each change is recorded with its reason, and listeners are notified.
- **DesireSet:** strategic goals such as MASTER_WEAK_CODING_TOPIC, ACHIEVE_INTERVIEW_READINESS and SURVIVE_IMMINENT_DRIVE. Each desire has a trigger, a satisfaction test and a priority computed from current beliefs.
- **IntentionStack:** the plan the agent has committed to, with a step cursor. Plans are procedural lists of steps, such as PLAN_CodingRemediation (7 steps) and PLAN_EmergencyCram (6 steps).

3. Deliberation cycle

1. Revise beliefs from percepts.
2. Recompute desire priorities and activate the highest-priority desire.
3. If the active desire has a different plan from the current intention, drop the current intention and commit to the new plan.
4. Execute one step, then repeat.

4. Implementation

```
class BeliefBase:
    def __init__(self):
        self._b: Dict[str, Any] = {}
        self._listeners: List[Callable] = []
        self.revision_count = 0

    def add(self, key, value):
        self._b[key] = value

    def get(self, key, default=None):
        return self._b.get(key, default)

    def update(self, key, value, reason="percept"):
        old = self._b.get(key)
        if old == value:
            return False
        self._b[key] = value
        self.revision_count += 1
        log("BELIEF", f"REVISION #{self.revision_count}: '{key}' {old!r} -&gt; {value!r}  ({reason})")
        for fn in self._listeners:
            fn(key, old, value)
        return True

    def on_change(self, fn):
        self._listeners.append(fn)

    def snapshot(self):
        return dict(self._b)

    def dump(self):
        print("    +-- BELIEF BASE " + "-" * 58)
```

```
        for k, v in self._b.items():
            print(f"    |   {k:&lt;26} = {v}")
        print("    +" + "-" * 73)

class IntentionStack:
    def __init__(self):
        self.stack: List[Intention] = []
        self.abandoned: List[Intention] = []

    def current(self):
        return self.stack[-1] if self.stack else None

    def commit(self, intention):
        self.stack.append(intention)
        log("INTENTION", f"COMMIT plan '{intention.plan_name}' for desire '{intention.desire}' "
                         f"({len(intention.steps)} steps)")

    def drop_current(self, reason):
        if self.stack:
            i = self.stack.pop()
            i.dropped = True
            self.abandoned.append(i)
            log("INTENTION", f"DROP plan '{i.plan_name}' at step {i.progress()} -- {reason}")

    def dump(self):
        print("    +-- INTENTION STACK " + "-" * 54)
        if not self.stack:
            print("    |   (empty)")
        for i in reversed(self.stack):
            print(f"    |   {i.plan_name:&lt;32} {i.progress()} steps  desire='{i.desire}'")
        print("    +" + "-" * 73)

    def deliberate_and_reconsider(self):
        """Reconsider intentions against the CURRENT belief state."""
        active = self.D.deliberate(self.B)
        if not active:
            return None
        top = active[0]
        cur = self.I.current()

        if cur is None:
            plan, steps = PLAN_LIBRARY[top.name]
            self.I.commit(Intention(top.name, plan, list(steps)))
            return self.I.current()

        # --- DYNAMIC PLAN RECALCULATION ---
        if cur.desire != top.name:
            cur_desire = next(d for d in self.D.desires if d.name == cur.desire)
            if top.priority(self.B) &gt; cur_desire.priority(self.B):
                log("RECALC", f"priority inversion detected: '{top.name}' "
                              f"({top.priority(self.B)}) &gt; '{cur.desire}' "
                              f"({cur_desire.priority(self.B)})")
                self.I.drop_current("superseded by higher-priority desire")
                plan, steps = PLAN_LIBRARY[top.name]
                self.I.commit(Intention(top.name, plan, list(steps)))
        # --- intention no longer needed (its desire got satisfied) ---
        elif cur_desire_satisfied(self.D, cur.desire, self.B):
            self.I.drop_current("desire already satisfied")
        return self.I.current()
```

5. Result

The agent first commits to PLAN_CodingRemediation (priority 10.24). Partway through, the placement cell reports a drive in 3 days instead of 45 (belief revision). SURVIVE_IMMINENT_DRIVE rises to priority 30.0, the agent drops the coding plan at step 3 of 7, and commits to PLAN_EmergencyCram with no human instruction.

Execution output (log file)

Captured from **logs/4.1_bdi_agent.log** (python3 bdi_agent.py). Timestamps and some values differ between runs; this is the recorded run.

```
# Activity 4.1_bdi_agent
# command: python3 bdi_agent.py
# exit code: 0


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```

```
 BDI GOAL-DRIVEN PLACEMENT PREPARATION AGENT (Assignment 4.1)
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================================================================================
 PHASE 0 :: INITIAL BDI STATE
================================================================================
    +-- BELIEF BASE -------------------------------------------------------
    |   coding_score               = 42
    |   aptitude_score             = 81
    |   interview_rating           = 7
    |   cgpa                       = 7.6
    |   days_to_drive              = 45
    |   weakest_topic              = Dynamic Programming
    |   projects_on_resume         = 2
    |   battery_of_mock_interviews_done = 3
    +----------------------------------------------------------------------
    +-- DESIRE SET --------------------------------------------------------
    |   SURVIVE_IMMINENT_DRIVE             INACTIVE   prio=20.0
    |   MASTER_WEAK_CODING_TOPIC           INACTIVE   prio=10.24
    |   CLEAR_APTITUDE_CUTOFF              INACTIVE   prio=7.0
    |   ACHIEVE_INTERVIEW_READINESS        INACTIVE   prio=6.5
    |   BUILD_PROJECT_PORTFOLIO            INACTIVE   prio=4.0
    +----------------------------------------------------------------------
    +-- INTENTION STACK ---------------------------------------------------
    |   (empty)
    +----------------------------------------------------------------------

================================================================================
 PHASE 1 :: DELIBERATION -&gt; INTENTION SELECTION
================================================================================
[12:26:32] [DESIRE      ] 'MASTER_WEAK_CODING_TOPIC' -&gt; ACTIVE (priority 10.24)
[12:26:32] [DESIRE      ] 'CLEAR_APTITUDE_CUTOFF' -&gt; ACHIEVED
[12:26:32] [DESIRE      ] 'ACHIEVE_INTERVIEW_READINESS' -&gt; ACTIVE (priority 6.5)
[12:26:32] [DESIRE      ] 'BUILD_PROJECT_PORTFOLIO' -&gt; ACTIVE (priority 4.0)
[12:26:32] [INTENTION   ] COMMIT plan 'PLAN_CodingRemediation' for desire 'MASTER_WEAK_CODING_TOPIC' (7 steps)
    +-- INTENTION STACK ---------------------------------------------------
    |   PLAN_CodingRemediation           0/7 steps  desire='MASTER_WEAK_CODING_TOPIC'
    +----------------------------------------------------------------------

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 PHASE 2 :: NORMAL PLAN EXECUTION (3 steps)
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[12:26:32] [EXECUTE     ] [PLAN_CodingRemediation 1/7] load weakest topic from belief base
[12:26:32] [EXECUTE     ] [PLAN_CodingRemediation 2/7] fetch 10 curated problems for that topic
[12:26:32] [EXECUTE     ] [PLAN_CodingRemediation 3/7] solve 4 EASY problems

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 PHASE 3 :: UNEXPECTED ENVIRONMENTAL BELIEF UPDATE (mid-execution)
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  &gt;&gt; External percept arrives: Amazon has moved its campus drive forward.

[12:26:32] [BELIEF      ] REVISION #1: 'days_to_drive' 45 -&gt; 3  (placement-cell notification)
[12:26:32] [BELIEF      ] REVISION #2: 'company_announced' None -&gt; 'Amazon'  (placement-cell notification)

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 PHASE 4 :: DYNAMIC PLAN RECALCULATION TRIGGERED BY BELIEF CHANGE
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[12:26:32] [DESIRE      ] 'SURVIVE_IMMINENT_DRIVE' -&gt; ACTIVE (priority 30.0)
[12:26:32] [RECALC      ] priority inversion detected: 'SURVIVE_IMMINENT_DRIVE' (30.0) &gt; 'MASTER_WEAK_CODING_TOPIC' (10.
[12:26:32] [INTENTION   ] DROP plan 'PLAN_CodingRemediation' at step 3/7 -- superseded by higher-priority desire
[12:26:32] [INTENTION   ] COMMIT plan 'PLAN_EmergencyCram' for desire 'SURVIVE_IMMINENT_DRIVE' (6 steps)
    +-- DESIRE SET --------------------------------------------------------
    |   SURVIVE_IMMINENT_DRIVE             ACTIVE     prio=30.0
    |   MASTER_WEAK_CODING_TOPIC           ACTIVE     prio=10.24
    |   CLEAR_APTITUDE_CUTOFF              ACHIEVED   prio=7.0
    |   ACHIEVE_INTERVIEW_READINESS        ACTIVE     prio=6.5
    |   BUILD_PROJECT_PORTFOLIO            ACTIVE     prio=4.0
    +----------------------------------------------------------------------
    +-- INTENTION STACK ---------------------------------------------------
    |   PLAN_EmergencyCram               0/6 steps  desire='SURVIVE_IMMINENT_DRIVE'
    +----------------------------------------------------------------------

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 PHASE 5 :: EXECUTING THE NEW (SHIFTED) INTENTION

... (middle of run omitted; full log in logs/4.1_bdi_agent.log) ...

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 VERIFICATION SUMMARY
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   Belief revisions performed  : 4
   Intentions abandoned        : 1
      - 'PLAN_CodingRemediation' dropped at 3/7 (desire 'MASTER_WEAK_CODING_TOPIC')
   Active intention now        : PLAN_EmergencyCram

   &gt;&gt; INTENTION SHIFT DEMONSTRATED: a belief change (days_to_drive 45 -&gt; 3)
      caused the agent to abandon PLAN_CodingRemediation mid-execution and
      re-commit to PLAN_EmergencyCram WITHOUT any human instruction.
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```

6. Viva points

Why separate beliefs, desires and intentions? A desire can change while an intention is still in progress; keeping them apart lets the agent drop the plan without losing the goal.

This is the clearest evidence in the project that the agent is not a fixed script: it reconsiders a commitment in the middle of execution.

7. Conclusion

The BDI cycle lets the agent respond to a changed environment by re-planning, which a reflex agent cannot do.